

Note 6: QEC2

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Date: 2026.5.6

6.1 Summary of Progress

1. 1
2. 1
3. 1

6.2 Further study of CSS code

$$\left\{ \begin{array}{c} \text{surface} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{topological} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{CSS} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{stabilizer} \\ \text{code} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{Quantum} \\ \text{code} \end{array} \right\}$$

Theorem 1 Let k be a field with $\text{char } k = 0$ and

1. $\mathcal{H}$ be a vector space over k ;
2. G be a finite group of $\text{GL}(\mathcal{H})$;
3. The subspace

$$\mathcal{H}^G := \{v \in \mathcal{H} \mid \forall g \in G, gv = v\};$$

then the operator

$$P_G := \frac{1}{|G|} \sum_{g \in G} g : \mathcal{H} \rightarrow \mathcal{H}$$

is a projector with $\mathcal{H}^G = \text{Im } P_G$.

Lemma 2 Let C be a linear code, then

$$\frac{1}{|C|} \sum_{y \in C} (-1)^{x \cdot y} = \chi_{C^\perp}(x)$$

Lemma 3 For $a, b \in \mathbb{F}_2^n$, denote $X^a = X^{a_1} \otimes \dots \otimes X^{a_n}$, $Z^b := Z^{b_1} \otimes \dots \otimes Z^{b_n}$, then we have

$$X^a |x\rangle = |x + a\rangle, \quad Z^b |y\rangle = (-1)^{b \cdot y} |y\rangle.$$

Lemma 4 Let H be the Hadamard matrix and $H_n := H^{\otimes n}$, $a, b, w \in \mathbb{F}_2^n$, then

- $H^2 = I$, $H^\dagger = H = H^{-1}$;
- $HX = ZH$, $HZ = XH$;
- $H_n^2 = I$, $H_n^\dagger = H_n = H_n^{-1}$;
- $H_n X^a = Z^a H_n$, $H_n Z^b = X^b H_n$.

Proof Recall the single-qubit Hadamard matrix and Pauli matrices:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

First, calculate H^2 :

$$H^2 = \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right) \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right) = \frac{1}{2} \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Since H is real symmetric, $H^\dagger = H$. Combined with $H^2 = I$, we have $H^{-1} = H$.

Next, verify $HX = ZH$:

$$HX = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0+1 & 1+0 \\ 0-1 & 1+0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

$$ZH = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1+0 & 1+0 \\ 0-1 & 0+1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

For $H_n = H^{\otimes n}$ (n-qubit Hadamard), we have:

$$H_n^2 = (H^{\otimes n})^2 = H^2 \otimes H^2 \otimes \cdots \otimes H^2 = I \otimes I \otimes \cdots \otimes I = I.$$

Since H_n is real symmetric, $H_n^\dagger = H_n$, so $H_n^{-1} = H_n$.

Let $a = (a_1, a_2, \dots, a_n) \in \mathbb{F}_2^n$, where $a_i \in \{0, 1\}$. From the single-qubit case $HX^{a_i} = Z^{a_i}H$, we have:

$$\begin{aligned} H_n X^a &= (H \otimes H \otimes \cdots \otimes H) \cdot (X^{a_1} \otimes X^{a_2} \otimes \cdots \otimes X^{a_n}) \\ &= (HX^{a_1}) \otimes (HX^{a_2}) \otimes \cdots \otimes (HX^{a_n}) \\ &= (Z^{a_1}H) \otimes (Z^{a_2}H) \otimes \cdots \otimes (Z^{a_n}H) \\ &= (Z^{a_1} \otimes Z^{a_2} \otimes \cdots \otimes Z^{a_n}) \cdot (H \otimes H \otimes \cdots \otimes H) \\ &= Z^a H_n. \end{aligned}$$

Similarly, $H_n Z^b = X^b H_n$ for any $b \in \mathbb{F}_2^n$. ■

Corollary 5 Let $|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{\epsilon \in \mathbb{F}_2^n} |\epsilon\rangle$ be the uniform superposition state. For any $w \in \mathbb{F}_2^n$,

$$H^{\otimes n} |w\rangle = Z^w |+\rangle^{\otimes n}.$$

Proof By the Hadamard lemma,

$$H^{\otimes n} |w\rangle = \frac{1}{\sqrt{2^n}} \sum_{\epsilon \in \mathbb{F}_2^n} (-1)^{w \cdot \epsilon} |\epsilon\rangle.$$

By the PauliZ lemma, $Z^w |\epsilon\rangle = (-1)^{w \cdot \epsilon} |\epsilon\rangle$, so

$$H^{\otimes n} |w\rangle = \frac{1}{\sqrt{2^n}} \sum_{\epsilon} Z^w |\epsilon\rangle = Z^w \left(\frac{1}{\sqrt{2^n}} \sum_{\epsilon} |\epsilon\rangle \right) = Z^w |+\rangle^{\otimes n}.$$

■

Error (noise) for CSS code

We encode $|x + C_2\rangle$ by

$$|\psi\rangle = |x + C_2\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} |x + y\rangle$$

Affected by noise $\mathcal{E} = X^{e_1} Z^{e_2}$ with $e_1, e_2 \in \mathbb{F}_2^n$, the state is changed to

$$|\psi_0\rangle = \mathcal{E}|\psi\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{(x+y) \cdot e_2} |x + y + e_1\rangle.$$

Recover operation for CSS code

The recover process can be expressed by the chain

$$\begin{aligned} |\psi_0\rangle &\mapsto |\psi_0\rangle \otimes |\vec{0}\rangle \xrightarrow{U_{H1}} |\psi_1\rangle \xrightarrow{\text{tr}_{\mathcal{H}}} |\psi_2\rangle \xrightarrow{X^{e_1}} |\psi_3\rangle \xrightarrow{H^{\otimes n}} |\psi_4\rangle \\ &\mapsto |\psi_4\rangle \otimes |\vec{0}\rangle \xrightarrow{U_{H2}} |\psi_5\rangle \xrightarrow{\text{tr}_{\mathcal{H}}} |\psi_6\rangle \xrightarrow{X^{e_2}} |\psi_7\rangle \xrightarrow{H^{\otimes n}} |\psi\rangle \end{aligned}$$

Where U_{H1}, U_{H2} is given by the lemma:

Lemma 6 Given $H \in \mathbb{F}_2^{r \times n}$, there is a unitary operation U_H such that

$$U_H(|c\rangle \otimes |0^{\otimes r}\rangle) = |c\rangle \otimes |Hc\rangle, \quad \forall c \in \mathbb{F}_2^n.$$

Proof Let $\mathcal{H}_A = (\mathbb{C}^2)^{\otimes n}$ and $\mathcal{H}_B = (\mathbb{C}^2)^{\otimes r}$. The total Hilbert space is $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, whose standard orthonormal basis is

$$\{|c\rangle \otimes |d\rangle \mid c \in \mathbb{F}_2^n, d \in \mathbb{F}_2^r\}.$$

We define the linear operator U_H on the standard basis of the whole Hilbert space as

$$U_H(|c\rangle \otimes |d\rangle) = |c\rangle \otimes |d + Hc\rangle,$$

It suffices to show this map is a permutation of all basis states. Suppose

$$|c_1\rangle \otimes |d_1 + Hc_1\rangle = |c_2\rangle \otimes |d_2 + Hc_2\rangle.$$

Tensor product basis vectors are linearly independent, so we have $c_1 = c_2$ and $d_1 + Hc_1 = d_2 + Hc_2$.

Since $c_1 = c_2$, we deduce $d_1 = d_2$. Hence the mapping is injective. The collection of all basis vectors is a finite set, so injectivity implies surjectivity.

Therefore U_H is bijective on the full basis set, which means it is a permutation of orthonormal bases. ■

Remark 7

$$\text{tr}_A U_H(- \otimes |0\rangle) = I_A$$

Theorem 8 Let $\mathcal{E} = X^{e_1} Z^{e_2}$ and

$$\mathcal{R} = H^{\otimes n} X^{e_2} \text{tr}_A U_{H2}(- \otimes |0\rangle) H^{\otimes n} X^{e_1} \text{tr}_A U_{H1}(- \otimes |0\rangle),$$

then $\mathcal{R} \circ \mathcal{E}(|\psi\rangle) = |\psi\rangle$.

Proof

$$\begin{aligned}
\mathcal{R} \circ \mathcal{E}(|\psi\rangle) &= H^{\otimes n} X^{e_2} \text{tr}_A U_{H_2}(- \otimes |0\rangle) H^{\otimes n} X^{e_1} \text{tr}_A U_{H_1}(- \otimes |0\rangle) (X^{e_1} Z^{e_2} |\psi\rangle) \\
&= H^{\otimes n} X^{e_2} \text{tr}_A U_{H_2}(- \otimes |0\rangle) H^{\otimes n} X^{e_1} X^{e_1} Z^{e_2} |\psi\rangle \\
&= H^{\otimes n} X^{e_2} \text{tr}_A U_{H_2}(- \otimes |0\rangle) H^{\otimes n} Z^{e_2} |\psi\rangle \\
&= H^{\otimes n} X^{e_2} \text{tr}_A U_{H_2}(- \otimes |0\rangle) X^{e_2} H^{\otimes n} |\psi\rangle \\
&= H^{\otimes n} X^{e_2} X^{e_2} H^{\otimes n} |\psi\rangle \\
&= H^{\otimes n} H^{\otimes n} |\psi\rangle \\
&= |\psi\rangle
\end{aligned}$$

■

Quantum Circuit for recover operation

Let $H = \sum_{i=1}^r \sum_{j=1}^n h_{ij} E_{ij}$, then

$$\begin{aligned}
U_{H(i,j)} (|c\rangle \otimes |d\rangle) &= |c\rangle \otimes |d + h_{ij} E_{ij} c\rangle \\
U_{H(1,2)} U_{H(1,1)} (|c\rangle \otimes |d\rangle) &= U_{H(1,2)} (|c\rangle \otimes |d + h_{11} E_{11} c\rangle) \\
&= |c\rangle \otimes |(d + h_{11} E_{11} c) + h_{12} E_{12} c\rangle
\end{aligned}$$

by induction

$$\begin{aligned}
\prod_{i=1}^r \prod_{j=1}^n U_{H(i,j)} (|c\rangle \otimes |d\rangle) &= |c\rangle \otimes \left| d + \sum_{i=1}^r \sum_{j=1}^n h_{ij} E_{ij} c \right\rangle \\
&= |c\rangle \otimes |d + Hc\rangle
\end{aligned}$$

So

$$\prod_{i=1}^r \prod_{j=1}^n U_{H(i,j)} = U_H.$$

Moreover,

6.3 Topological code

Proposition 9 Let $C_2 \subseteq C_1 \subseteq \mathbb{F}_2^n$, and let H_Z, H_X be the paritycheck matrix of $C_1^\perp, C_2$, then the CSS code is the stabilizer code with stabilizer group

$$\mathcal{G} = \{X^a Z^b \mid H_X a = \mathbf{0}, H_Z b = \mathbf{0}\} = \{X^a Z^b \mid (a, b) \in C_2 \times C_1^\perp\},$$

Lemma 10 If $H_X \in \mathbb{F}_2^{m_X \times n}, H_Z \in \mathbb{F}_2^{m_Z \times n}$ with $m_X, m_Z \leq n$, then

$$H_X H_Z^T = 0 \iff C_Z^\perp \subseteq C_X.$$

Where $C_X = \text{Ker} H_X, C_Z = \text{Ker} H_Z$.

Proof : $H_X H_Z^T = 0 \iff \text{Im} H_Z^T \subseteq \text{Ker} H_X = C_X$. It suffices to prove $\text{Im} H_Z^T = C_Z^\perp$.

- Consider the linear map

$$A : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n = \mathbb{F}_2^{m_Z} \oplus \mathbb{F}_2^{n-m_Z}, \quad x \mapsto \begin{pmatrix} H_Z x \\ 0 \end{pmatrix} = \begin{pmatrix} H_Z \\ 0 \end{pmatrix} x;$$

then

$$A = \begin{pmatrix} H_Z \\ 0 \end{pmatrix}, \quad A^T = (H_Z^T \ 0).$$

- The adjoint map is

$$A^* : \mathbb{F}_2^n = \mathbb{F}_2^{m_Z} \oplus \mathbb{F}_2^{n-m_Z} \rightarrow \mathbb{F}_2^n, \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto (H_Z^T \ 0) \begin{pmatrix} x \\ y \end{pmatrix} = H_Z^T x$$

- Using the lemma, we have

$$\begin{aligned} \text{Im} H_Z^T &= \text{Im} A^* = (\text{Ker} A)^\perp \\ &= \left\{ x \in \mathbb{F}_2^n \mid \begin{pmatrix} H_Z x \\ 0 \end{pmatrix} = 0 \right\}^\perp \\ &= (\text{Ker} H_Z)^\perp = C_Z^\perp \end{aligned}$$

The proof is complete. ■

Remark 11

$$\begin{aligned} \{X^a \mid a \in C_X\} &\leftrightarrow C_1 = C_X = \text{Ker} H_X \\ \{Z^b \mid a \in C_Z\} &\leftrightarrow C_2 = C_Z^\perp = (\text{Ker} H_Z)^\perp = \text{Im} H_Z^T \\ C_2 \subseteq C_1 &\leftrightarrow H_X H_Z^T = 0 \leftrightarrow H_Z H_X^T = 0 \end{aligned}$$

In other word,

$$x + C_2 \in C_1 / C_2 = \text{Ker} H_X / \text{Im} H_Z^T = H_1(K)$$

belong to homological group of K , where K denote the chain complex

$$\mathbb{F}_2^{m_Z} \xrightarrow{H_Z^T} \mathbb{F}_2^n \xrightarrow{H_X} \mathbb{F}_2^{m_X}.$$

In general, if you give me a chain complex

$$C_2(K; \mathbb{F}_2) \xrightarrow{\partial_2} C_1(K; \mathbb{F}_2) \xrightarrow{\partial_1} C_0(K; \mathbb{F}_2),$$

i.e. $\partial_1 \circ \partial_2 = 0$. then

$$H_X = \partial_1, H_Z = \partial_2^T$$

define a CSS code.

6.4 Plan for next week

6.5 Appendix

6.5.1 GF(4) check for symplectic product

Lemma 12 *We have the following isomorphisms as field GF(4)*

$$\mathbb{F}_4 \simeq \frac{\mathbb{F}_2[x]}{\langle x^2 + x + 1 \rangle} \simeq \mathbb{F}_2[\omega] = \{a + b\omega : a, b \in \mathbb{F}_2\}$$

while $\omega \neq 0, 1$ and $\omega^2 = \omega + 1$. So the field is unique up to isomorphism.

Proof Since $\mathbb{F}_4/\mathbb{F}_2$ is a field extension with $[\mathbb{F}_4 : \mathbb{F}_2] = 2$, there is $\omega \in \mathbb{F}_4$ such that

$$\mathbb{F}_4 = \mathbb{F}_2[\omega] = \{a + b\omega : a, b \in \mathbb{F}_2\}.$$

Then

$$\mathbb{F}_4 = \mathbb{F}_2[\omega] = \{0, 1, \omega, \omega^2\} = \{0, 1, \omega, \omega + 1\},$$

Thus $\omega^2 = \omega + 1$. Consider the ring homomorphism

$$\phi : \mathbb{F}_2[x] \rightarrow \mathbb{F}_4, \quad f(x) \mapsto f(\omega),$$

We have $\mathbb{F}_4 = \text{im } \phi \simeq \mathbb{F}_2[x] / \ker \phi$. And for $f \in \ker \phi$. There is $q(x), r(x) \in \mathbb{F}_2[x]$

$$f(x) = (x^2 + x + 1)q(x) + r(x), \quad \deg r(x) < 2.$$

$$0 = f(\omega) = (\omega^2 + \omega + 1)q(\omega) + r(\omega) = r(\omega)$$

So we have $r(x) = 0 \implies \ker \phi = \langle x^2 + x + 1 \rangle$, a maximal ideal of $\mathbb{F}_2[x]$. ■

Corollary 13 $\omega \mapsto \omega^2 = \omega + 1$ induces a field automorphism of $\mathbb{F}_4$.

Definition 14 For $u = a + b\omega, a, b \in \mathbb{F}_2$, define

1. Conjugate element

$$\bar{u} := a + b\omega^2 = a + b + b\omega.$$

2. Trace

$$\text{tr}(u) := \text{Tr}_{\mathbb{F}_2}(u) := u + \bar{u} = b$$

Remark 15 $u + \bar{u} = a + b\omega + a + b\omega^2 = b(\omega + \omega^2) = b$.

Proposition 16 If $(a, b), (c, d) \in \mathbb{F}_2^2, u = a + b\omega, v = c + d\omega$, then

$$\text{tr}(u\bar{v}) = \text{tr}(\bar{u}v) = ad + bc.$$

Proof Proved by formula $\text{tr}(a_0 + b_0\omega) = b_0$ and

$$\begin{aligned} u\bar{v} &= (a + b\omega)(c + d\omega^2) \\ &= ac + ad\omega^2 + bc\omega + bd \\ &= ac + bd + ad(\omega + 1) + bc\omega \\ &= ac + bd + ad + (ad + bc)\omega \end{aligned}$$
■

Corollary 17 *Let the correspondence be*

$$I \leftrightarrow 0, \quad X \leftrightarrow 1, \quad Z \leftrightarrow \omega, \quad Y \leftrightarrow \omega^2.$$

For Pauli operators

$$P = P_1 \otimes P_2 \otimes \cdots \otimes P_n, \quad Q = Q_1 \otimes Q_2 \otimes \cdots \otimes Q_n,$$

let $P_j \leftrightarrow u_j$ and $Q_j \leftrightarrow v_j$ in $\mathbb{F}_4$. Then

$$[P, Q] = 0 \iff \sum_{j=1}^n \text{tr}(u_j \bar{v}_j) = 0.$$

Proof We have known that if $P = X^a Z^b, Q = X^c Z^d$, then

$$[P, Q] = 0 \iff a \cdot d + b \cdot c = 0 \iff \sum_{j=1}^n a_j d_j + b_j c_j = 0.$$

which is equivalent to $u_j = a_j + b_j \omega, v_j = c_j + d_j \omega$ and

$$\sum_{j=1}^n \text{tr}(u_j \bar{v}_j) = 0.$$

This completes the proof. ■

6.5.2 Some supplementary of materials for linear algebra

Lemma 18 *Consider finite dimensional vector space V over a field k . Given a **non-degenerate** bilinear form $B : V \times V \rightarrow k$ a non-, then for any linear map $A : V \rightarrow V$, we have a unique linear map $A^* : V \rightarrow V$ such that*

$$B(Av, w) = B(v, A^*w) : \forall v, w \in V.$$

Remark 19 . We say A^* is the adjoint map of A .

Theorem 20 *For any linear map $A : V \rightarrow V$, we have*

$$(\text{Im} A)^\perp = \text{Ker} A^*, \quad (\text{Ker} A)^\perp = \text{Im} A^*;$$

$$V = \text{Ker} A^* \oplus^\perp \text{Im} A = \text{Ker} A \oplus^\perp \text{Im} A^*;$$

where $W^\perp := \{v \in V | \forall w \in W, B(v, w) = 0\}$.

Proof For $B : V \times V \rightarrow k$ is non-degenerate, we have $B(V, w) = \{0\} \implies w = 0$. So

$$\begin{aligned} (\text{Im} A)^\perp &= \{v \in V | \forall w \in V, B(Aw, v) = 0\} \\ &= \{v \in V | \forall w \in V, B(w, A^*v) = 0\} \\ &= \{v \in V | A^*v = 0\} \\ &= \text{Ker} A^*. \end{aligned}$$

Similarly we can prove that $(\text{Ker} A)^\perp = \text{Im} A^*$. ■

Corollary 21 If $A^* = A$ or $A^* = -A$, then

$$V = \text{Ker} A \oplus^\perp \text{Im} A.$$

Theorem 22 The following statements are equivalent:

1. $V = \text{Ker} A \oplus \text{Im} A$;
2. $V = \text{Ker} A + \text{Im} A$;
3. $\text{Ker} A \cap \text{Im} A = \{0\}$;
4. $\text{Ker} A = \text{Ker} A^2$;
5. $\text{Im} A = \text{Im} A^2$.
6. $\text{Im} A = W, A|_W : W \rightarrow W$ is invertible.

Proof

- $1 \Rightarrow 2$ is obvious.
- $2 \Rightarrow 3$: By dimension formulas

$$\dim V = \dim \text{Ker} A + \dim \text{Im} A$$

and

$$\dim (\text{Ker} A + \text{Im} A) = \dim \text{Ker} A + \dim \text{Im} A - \dim (\text{Ker} A \cap \text{Im} A),$$

we have $\dim (\text{Ker} A \cap \text{Im} A) = 0$.

- $3 \Rightarrow 4$: $\text{Ker} A \subseteq \text{Ker} A^2$ is obvious.
If $A^2 x = 0$, then $Ax \in \text{Ker} A \cap \text{Im} A = \{0\}$. So $Ax = 0 \implies x \in \text{Ker} A$.
- $4 \Rightarrow 5$: $\text{Im} A^2 \subseteq \text{Im} A$ is obvious. For the dimension formulas

$$\begin{aligned} \dim V &= \dim \text{Im} A + \dim \text{Ker} A, \\ \dim V &= \dim \text{Im} A^2 + \dim \text{Ker} A^2, \end{aligned}$$

we have $\dim \text{Im} A^2 = \dim \text{Im} A$.

- Since $\text{Im} A = \text{Im} A^2 = W$, so $A|_W : W \rightarrow W$ is surjective, so is injective for $\dim V < \infty$.
- $6 \Rightarrow 5$: For $A|_W : W \rightarrow W$ is invertible, $\dim AW = \dim W$. So

$$\dim \text{Im} A^2 = \dim A^2 V = \dim AW = \dim W = \dim \text{Im} A.$$

- $5 \Rightarrow 4$: Similar to $4 \Rightarrow 5$.
- $5, 4 \Rightarrow 3$: If $x \in \text{Ker} A \cap \text{Im} A$, then

$$Ax = 0, x = Ay \implies A^2 y = 0 \implies y \in \text{Ker} A^2 = \text{Ker} A \implies x = Ay = 0.$$

- $3 \Rightarrow 1$: By formula

$$\dim (\text{Ker} A + \text{Im} A) = \dim \text{Ker} A + \dim \text{Im} A - \dim (\text{Ker} A \cap \text{Im} A) = \dim V - 0 = \dim V.$$

■

Corollary 23 If $A^* = A$ or $A^* = -A$, then A satisfies all the equivalent conditions above.

6.5.3 Simplicial Complexes and Simplicial Homology Groups over $\mathbb{F}_2$

Definition 24 (Simplex and Face) An n -simplex $\underline{s} = (a_0, a_1, \dots, a_n)$ is the convex hull of $n + 1$ affinely independent points $a_0, \dots, a_n$ in Euclidean space, called *vertices*. A simplex $\underline{t}$ is a **face** of $\underline{s}$ ($\underline{t} < \underline{s}$) if all vertices of $\underline{t}$ are vertices of $\underline{s}$.

Definition 25 (Boundary of a Simplex) The **boundary** $\partial \underline{s}$ of an n -simplex $\underline{s}$ is the union of all its proper faces. Geometrically, $\partial \underline{s} \cong S^{n-1}$.

Definition 26 (Oriented Simplex) Let $\underline{s} = (a_0, a_1, \dots, a_n)$ be an n -simplex. An *oriented n -simplex* s is defined as

$$s = [a_0, a_1, \dots, a_n], \quad -s = [a_{\sigma(0)}, a_{\sigma(1)}, \dots, a_{\sigma(n)}]$$

where σ is an **odd permutation** of indices. Two orderings define the same orientation if related by an even permutation.

Definition 27 (Simplicial Complex) A *simplicial complex* K is a set of simplices such that:

1. Any two simplices are either disjoint or intersect in a common face.
2. If $s \in K$, then all faces of s are also in K .

Definition 28 ($\mathbb{F}_2$ -Chain Group) The q -dimensional chain group $C_q(K; \mathbb{F}_2)$ is the $\mathbb{F}_2$ -vector space generated by all oriented q -simplices of K . We write $N_q = \dim_{\mathbb{F}_2} C_q(K; \mathbb{F}_2)$.

Remark 29 For $-1 = 1$ over $\mathbb{F}_2$, $-s = s$, so **orientation is irrelevant**.

Example 30 Define the simplicial complex

$$K = \{[a_0], [a_1], [a_2], [a_0, a_1], [a_0, a_2], [a_1, a_2], [a_0, a_1, a_2]\}.$$

1. The vector space $C_2(K; \mathbb{F}_2)$ is generated by the unique 2-simplex $[a_0, a_1, a_2]$, so its basis is $\{[a_0, a_1, a_2]\}$ and $N_2 = \dim_{\mathbb{F}_2} C_2(K; \mathbb{F}_2) = 1$.
2. The vector space $C_1(K; \mathbb{F}_2)$ is generated by the three 1-simplices $[a_1, a_2], [a_0, a_2], [a_0, a_1]$, so its basis is $\{[a_1, a_2], [a_0, a_2], [a_0, a_1]\}$ and $N_1 = \dim_{\mathbb{F}_2} C_1(K; \mathbb{F}_2) = 3$.

Definition 31 (Boundary Homomorphism over $\mathbb{F}_2$) The boundary homomorphism

$$\partial_q : C_q(K; \mathbb{F}_2) \rightarrow C_{q-1}(K; \mathbb{F}_2)$$

is defined on an oriented q -simplex $s = [a_0, a_1, \dots, a_q]$ by

$$\partial_q s := \sum_{i=0}^q (-1)^i [a_0, \dots, \hat{a}_i, \dots, a_q] = \sum_{i=0}^q [a_0, \dots, \hat{a}_i, \dots, a_q]$$

and extended linearly over $\mathbb{F}_2$.

Remark 32 Let $N_q = \dim_{\mathbb{F}_2} C_q(K; \mathbb{F}_2)$, $N_{q-1} = \dim_{\mathbb{F}_2} C_{q-1}(K; \mathbb{F}_2)$. For a general q -chain

$$c = \sum_{j=1}^{N_q} \lambda_j s_j \in C_q(K; \mathbb{F}_2)$$

with coefficients $\lambda_j \in \mathbb{F}_2$, the linear extension is given by

$$\partial_q c = \sum_{j=1}^{N_q} \lambda_j \partial_q s_j.$$

And we can get a chain of morphism

$$\dots \xrightarrow{\partial_{q+1}} C_q(K) \xrightarrow{\partial_q} C_{q-1}(K) \xrightarrow{\partial_{q-1}} \dots \xrightarrow{\partial_2} C_1(K) \xrightarrow{\partial_1} C_0(K).$$

Example 33 Define the simplicial complex

$$K = \{ [a_0], [a_1], [a_2], [a_0, a_1], [a_0, a_2], [a_1, a_2], [a_0, a_1, a_2] \}.$$

By the definition of boundary homomorphism over $\mathbb{F}_2$,

$$\partial_2[a_0, a_1, a_2] = \sum_{i=0}^2 [a_0, \dots, \hat{a}_i, \dots, a_2] = [a_1, a_2] + [a_0, a_2] + [a_0, a_1].$$

According to the matrix representation of boundary homomorphism, the boundary matrix $\partial_2 \in \mathbb{F}_2^{3 \times 1}$ is

$$\partial_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Represent the 2-chain $[a_0, a_1, a_2]$ as the vector $\mathbf{c} = (1) \in \mathbb{F}_2^1$, then

$$\partial_2[a_0, a_1, a_2] = \partial_2 \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Matrix Representation of Boundary Homomorphism is given by the proposition: Let $N_q = \dim_{\mathbb{F}_2} C_q(K; \mathbb{F}_2)$

Proposition 34 Suppose $C_q(K; \mathbb{F}_2)$ has basis $\{s_1, \dots, s_{N_q}\}$ and $C_{q-1}(K; \mathbb{F}_2)$ has basis $\{t_1, \dots, t_{N_{q-1}}\}$. The homomorphism ∂_q corresponds to a boundary matrix $M \in \mathbb{F}_2^{N_{q-1} \times N_q}$ with entries

$$M_{ij} = \begin{cases} 1 & t_i \text{ is contained in } \partial_q s_j, \\ 0 & \text{otherwise.} \end{cases}$$

For any chain $c \in C_q(K; \mathbb{F}_2)$ represented by a vector $\mathbf{c} \in \mathbb{F}_2^{N_q}$, we have

$$\partial_q c = M \mathbf{c}.$$

Proof Since $\{s_1, \dots, s_{N_q}\}$ is a basis of $C_q(K; \mathbb{F}_2)$, any q -chain can be uniquely written as

$$c = \sum_{j=1}^{N_q} c_j s_j, \quad \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_{N_q} \end{pmatrix} \in \mathbb{F}_2^{N_q}.$$

By linearity of ∂_q over $\mathbb{F}_2$,

$$\partial_q c = \sum_{j=1}^{N_q} c_j \partial_q s_j.$$

For each basis element $s_j \in C_q$, expand $\partial_q s_j$ in the basis $\{t_1, \dots, t_{N_{q-1}}\}$ of C_{q-1} :

$$\partial_q s_j = \sum_{i=1}^{N_{q-1}} M_{ij} t_i,$$

where $M_{ij} = 1$ if $t_i \in \partial_q s_j$, else $M_{ij} = 0$, by definition of the matrix entries. Substitute into the expression for $\partial_q c$:

$$\partial_q c = \sum_{j=1}^{N_q} c_j \left(\sum_{i=1}^{N_{q-1}} M_{ij} t_i \right) = \sum_{i=1}^{N_{q-1}} \left(\sum_{j=1}^{N_q} M_{ij} c_j \right) t_i.$$

The coefficient vector of this linear combination is exactly $M\mathbf{c} \in \mathbb{F}_2^{N_{q-1}}$. Therefore, $\partial_q c = M\mathbf{c}$. ■

Definition 35 ($\mathbb{F}_2$ -Homology Group) The q -dimensional homology group over $\mathbb{F}_2$ is

$$H_q(K; \mathbb{F}_2) = \ker \partial_q / \text{Im } \partial_{q+1}.$$

Theorem 36 (Homotopy Invariance of Homology Groups) Let K, L be finite simplicial complexes. If their geometric realizations $|K|$ and $|L|$ are homotopy equivalent, then for every integer $q \geq 0$, the $\mathbb{F}_2$ -homology groups are isomorphic:

$$H_q(K; \mathbb{F}_2) \cong H_q(L; \mathbb{F}_2).$$

Definition 37 (Betti Number over $\mathbb{F}_2$) The q -th Betti number over $\mathbb{F}_2$ is $\beta_q = \dim_{\mathbb{F}_2} H_q(K; \mathbb{F}_2)$.

Definition 38 (Euler Characteristic) Let K be a finite simplicial complex, $N_q = \dim_{\mathbb{F}_2} C_q(K; \mathbb{F}_2)$ the number of q -simplices in K . The Euler characteristic of K is defined as

$$\chi(K) = \sum_{q=0}^{\dim K} (-1)^q N_q.$$

Theorem 39 (Euler-Poincaré Formula & Homotopy Invariance) Let K be a finite simplicial complex, $\beta_q = \dim_{\mathbb{F}_2} H_q(K; \mathbb{F}_2)$ the q -th $\mathbb{F}_2$ -Betti number. Then

$$\chi(K) = \sum_{q=0}^{\dim K} (-1)^q \beta_q.$$

Moreover, Euler characteristic is a **homotopy invariant**.

Proof Let $n = \dim K$. By ranknullity theorem over $\mathbb{F}_2$,

$$N_q = \dim C_q = \dim \ker \partial_q + \dim \text{Im } \partial_q.$$

From the short exact sequence $0 \rightarrow \text{Im } \partial_{q+1} \rightarrow \ker \partial_q \rightarrow H_q \rightarrow 0$,

$$\dim \ker \partial_q = \dim \text{Im } \partial_{q+1} + \beta_q.$$

Substitute into the Euler characteristic:

$$\begin{aligned} \chi(K) &= \sum_{q=0}^n (-1)^q N_q = \sum_{q=0}^n (-1)^q (\dim \text{Im } \partial_{q+1} + \beta_q + \dim \text{Im } \partial_q) \\ &= \sum_{q=0}^n (-1)^q \beta_q + \sum_{q=0}^n (-1)^q \dim \text{Im } \partial_{q+1} + \sum_{q=0}^n (-1)^q \dim \text{Im } \partial_q. \end{aligned}$$

Reindexing the first image sum shows the two boundary image terms cancel exactly, hence

$$\chi(K) = \sum_{q=0}^n (-1)^q \beta_q.$$
■

References

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